

Jacob Beeson

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Molecular Bioengineering student with practical experience across Python development, data analysis, sensor systems, CAD prototyping, experimental testing and hardware-software integration. Experienced in developing technical projects from initial requirements through to testing and refinement, while working across engineering, research and user-focused environments.

UK and Irish passport holder.

EDUCATION

Imperial College London

Oct. 2023 – Expected Aug. 2027

MEng Molecular Bioengineering

London, UK

- Expected classification: First Class.
- Relevant areas: applied programming, data analysis, mathematical modelling, sensor technology, biomimetics and engineering design.

Barton Peveril Sixth Form College

2020 – 2022

A-Levels

Southampton, UK

- Mathematics: A*, Chemistry: A*, Biology: A, Further Mathematics: A (AS-Level).

ENGINEERING & RESEARCH EXPERIENCE

Research Intern – Magnetic Sensor Systems & Nanorobotics

Jun. 2026 – Present

DKFZ / TU Dresden

Dresden, Germany

- Develop and maintain Python-based data acquisition, signal generation, automation and analysis tools within a multidisciplinary nanorobotics research team.
- Design and conduct experimental workflows integrating magnetic actuation, sensor readout and signal analysis to investigate system behaviour and support reliable testing.

MRI-Compatible Knee Loading Device

Sept. 2025 – Jun. 2026

Group Design Project, Imperial College London

London, UK

- Co-engineered a mechanical system for controlling knee flexion and loading within a 3T MRI scanner, working within strict safety, material and spatial constraints.
- Led CAD prototyping and FEA validation in ANSYS alongside iterative user testing, improving structural performance, comfort and experimental repeatability.

Summer Research Fellow – Wearable Electrochemical Sensing

Jun. 2025 – Sept. 2025

Institute of Chemical Research of Catalonia (ICIQ)

Tarragona, Spain

- Built a Python data-processing pipeline that automated electrochemical sensor analysis, reducing manual processing time by over 50% and improving the reporting workflow.
- Designed, 3D-printed and iteratively tested a wearable sensor enclosure, addressing usability, integration and durability requirements through repeated prototype cycles.

ADDITIONAL PROJECTS & VENTURES

Mountain Rescue Wearable Device | Imperial CGCU Makeathon – 1st Place

Feb. 2026

- Won first place by engineering a wearable IoT safety device capable of detecting unconsciousness and automatically transmitting a distress signal.
- Developed GPS-based application functionality for real-time casualty localisation and route planning under a short project deadline.

Optoblock | Founder – product development & AI integration

Apr. 2026 – Present

- Founded a modular productivity application, leading development from requirements gathering and interface design through to a functioning, tested system with customisable draggable tiles.
- Designed and integrated a voice-driven feature that converts unstructured spoken input into organised tasks using speech recognition and natural-language processing.

Sansoperator | Founder – digital art & spatial design

Jan. 2024 – Present

- Founded an award-winning digital art and spatial design venture, managing more than 10 public installations using real-time rendering, projection mapping and interactive visual systems.

SKILLS

Programming & Data: Python, MATLAB, Linux, C++/Arduino, signal processing, data analysis, data pipelines, computer vision and GUI/application development.

Engineering Systems: Sensor integration, data acquisition, National Instruments **nidaqmx**, hardware-software integration, IoT systems and real-time data processing.

Design & Prototyping: SolidWorks, Fusion 360, ANSYS FEA, 3D printing, mechanical prototyping, microfabrication and iterative product development.

Testing & Research: Experimental design, system troubleshooting, test automation, verification, validation, technical documentation and analytical problem-solving.

Collaboration & Communication: Multidisciplinary teamwork, stakeholder-facing project work, technical presentations, workshop leadership and user-centred design.